Response to the Reviewers

Resubmission of the manuscript previously titled "Coordinated V2G Voltage and Frequency-State Regulation in a Two-Area System: A Gaussian Actor–Critic Simulation Framework" (manuscript ID: [previous ID])

New title: "Two-Time-Scale Coordination of Vehicle-to-Grid Voltage Support and Frequency Reserve in a Two-Area System Using Proximal Policy Optimization"

Dear Editor and Reviewers,

We thank both reviewers for their careful and direct assessment. They identified the central weakness of the previous version correctly: the controller was never trained, the network and frequency models were algebraic surrogates, and the comparison curves were derived rather than simulated. We agree with this assessment. Rather than revising the text, we rebuilt the study from the model upward. The main changes are:

1. The surrogate voltage equations were replaced by an AC power flow on the IEEE 33-bus feeder (Baran–Wu data) in each area, solved by a backward/forward sweep and verified against pandapower (maximum voltage difference 1.2×10^{-10} p.u.).
2. The algebraic frequency equation was replaced by a dynamic two-area load-frequency control model with inertia, damping, governor–turbine dynamics, a tie line with synchronizing coefficient, ACE-based AGC, and a saturated V2G droop loop, integrated at a 20-ms step. Hourly scheduling and second-level frequency dynamics are now separate time scales.
3. All actions and states have physical units: fleet size, battery capacity, charger rating, efficiencies, availability, driving energy, area base, and tie-line dynamics are specified (Table 1 of the manuscript). Sampled actions are clipped before use, and set-points are projected onto charger and energy limits; raw-action clippings and physical projections are counted separately.
4. PPO is now actually trained (Stable-Baselines3) with 5 independent seeds for 399 360 steps each, with a fixed validation set and a stated convergence criterion. All diagnostic or analytical curves of the previous version (old Figs. 3 to 8 and 13) were removed.
5. Three benchmark controllers are simulated independently under identical scenarios, disturbances, network, and constraints: uncontrolled charging, a conventional rule-based volt-watt and droop controller, and a day-ahead optimization with perfect foresight.
6. Results are reported on 30 held-out days and 20 out-of-distribution stress days with quantitative metrics, standard deviations across days and seeds, and paired significance tests. A reward ablation is included.
7. The energy exchanged by the frequency droop loop is included in the fleet energy balance with both efficiencies. Four treatments of this term were compared against exact 3600-s simulations of every evaluated hour; the adopted treatment tracks the exact value to within 6% and the results are insensitive to the choice (Section IV-E).
8. Raw-action saturation of the Gaussian policy is reported per action dimension and compared with a bounded (tanh-squashed, gSDE) policy trained at the same budget (Section IV-F).
9. All policies were trained from scratch in the final model version; every number in the manuscript is recomputed from the raw result files by an independent verification script, and the code, trained policies, data, checksums, and exact package versions are released.

Because the model, the method, and all results changed, we are submitting the work as a new manuscript and summarize below how each comment was addressed. Section, table, and figure numbers refer to the new manuscript.

Reviewer 1

Comment 1.1

Implement and document a complete training procedure, with genuine training and held-out evaluation results, or redefine the contribution.

Response: We implemented the complete PPO procedure (clipped surrogate objective, generalized advantage estimation, minibatch updates) with Stable-Baselines3. Training uses 5 seeds, four parallel environments, and randomized daily scenarios. Every 6144 steps the deterministic policy is evaluated on ten fixed validation days, and the policy is finally tested on 30 held-out days and 20 stress days that are never used for training or model selection. The final validation cost changes by at most 1.0% over the last 50 000 steps.

Changes in the manuscript: Section III-C (training procedure), Table 1 (hyperparameters), Fig. 4 (learning curves on training and validation days), Section IV-B.

Comment 1.2

Implement a documented network model with branch parameters, nodal injections, and a reproducible power-flow solution, or stop presenting the surrogate as IEEE 33-bus.

Response: Each area now contains the IEEE 33-bus feeder with the original branch impedances and nominal loads (12.66 kV), taken from the pandapower case library. Net nodal injections combine load, PV, and V2G power at specified buses, and the power flow is solved by the direct BIBC/BCBV method. The base case reproduces the published values (minimum voltage 0.9131 p.u. at bus 18, loss 202.68 kW), and the solver agrees with pandapower Newton–Raphson to within 1.2×10^{-10} p.u. over 20 random operating points.

Changes in the manuscript: Section II-B, Eqs. (1) and (2), Fig. 2, Section IV-A.

Comment 1.3

Provide a physically justified frequency model, define the units of the coefficient 0.15, and relate actions to power imbalance.

Response: The coefficient 0.15 and the algebraic frequency equation were removed. The new model (Eqs. (9) to (14)) is the standard two-area LFC model in per unit of a 500-MW area base and 50 Hz, with inertia constants that decrease with PV output, load damping, governor and turbine time constants, droop, a tie line with synchronizing coefficient T_{12}, ACE, and integral AGC. The V2G action enters physically: the reserve in MW sets the droop gain and the saturation of the fleet's response. The model reproduces the closed-form steady-state deviation (-0.02381 Hz simulated against -0.02381 Hz theoretical) and tie-line change.

Changes in the manuscript: Section II-E, Table 1, Section IV-A.

Comment 1.4

Simulate the controlled and baseline cases independently under identical conditions, include an uncontrolled case and a conventional controller, with quantitative indices.

Response: All controllers now run in the same environment with the same scenarios, the same per-hour disturbance seeds, the same network, and the same constraints. The benchmarks are uncontrolled charging (UC), a rule-based volt-watt and droop controller (RB), and a day-ahead optimization with perfect foresight (DAO). Sixteen quantitative indices are reported with standard deviations, and paired Wilcoxon tests are given. On the test days, PPO reduces the daily cost by 11.0%, 44.8%, and 71.6% relative to DAO, RB, and UC ($p = 1.9 \times 10^{-9}$).

Changes in the manuscript: Sections III-D and III-E, Tables 2 and 9, Figs. 5 to 9.

Comment 1.5

The active-power-loss comparison in Fig. 13 cannot be reproduced.

Response: The figure has been removed. Feeder losses are now computed from the branch currents of the power-flow solution and are reported as one of the indices in Table 2, for every controller.

Changes in the manuscript: Table 2.

Comment 1.6

Define the physical units and scaling of each action, impose explicit bounds, and demonstrate compliance with fleet power, converter, and inter-area limits.

Response: The action now consists of a normalized set-point and a reserve share per area. Eqs. (17) and (18) map them to MW using the plugged-in charger rating, and the set-point is projected onto the charger and energy limits of Eqs. (4) and (5); the reserve is limited so that the droop deployment also respects these limits. Actions sampled from the Gaussian policy are clipped to $[-1, 1]$ before this mapping. Raw-action clipping and physical projection are counted separately: the PPO policy mean lies outside $[-1, 1]$ for 49.2% of action components (saturation at the design limits), while the physical projection is never active for PPO. The combined scheduled and droop power is audited at every 20-ms step and never exceeds the charger or energy limits (Section IV-E, Table 6). Tie-line power is a dynamic state of the LFC model, and its RMS deviation is reported. The "MW" labels of the old figures, which had no physical basis, are gone.

Changes in the manuscript: Section III-A, Section IV-E, Tables 2 and 6.

Comment 1.7

Provide an energy-consistent fleet model.

Response: The fleet model (Eqs. (3) to (7)) specifies 300 vehicles per feeder, 60-kWh batteries, 7.4-kW chargers, charging and discharging efficiencies of 0.92, a 1-h step, SoC limits of 0.20 and 0.90, time-varying plug-in availability, and 9 kWh of daily driving energy per vehicle. SoC-dependent power limits follow from the remaining energy. The energy exchanged by the frequency droop loop is computed from the fast-layer simulations of every hour, with both efficiencies, and added to the energy balance (Eq. (6)); four treatments of this term are compared with exact 3600-s simulations in Section IV-E and its size is reported in Table 6. Raw-action clippings, projection events, and unmet mobility energy are recorded and reported.

Changes in the manuscript: Section II-D, Tables 1, 2, and 6.

Comment 1.8

Justify the penalty weights, specify the voltage and frequency limits, penalize both lower and upper voltage deviations, and demonstrate the effect of the reward design.

Response: The reward now penalizes both undervoltage and overvoltage outside 0.95 to 1.05 p.u., the frequency nadir, the tie-line deviation, battery throughput, unmet mobility energy, and the end-of-day SoC deficit (Eq. (21)). The weights were set so that voltage and frequency costs are of the same order on an uncontrolled day, which is stated in the text. A reward ablation (Table 10) retrains the policy without the overvoltage term, without the frequency terms, and without the throughput term, and evaluates each with the full cost against a reference trained with the same budget. Removing the frequency terms clearly reduces the reserve (10.1 MW instead of 22.9 MW) and deepens the nadir, which confirms that the learned reserve allocation comes from these terms. Removing the overvoltage or throughput term gives no difference beyond the seed-to-seed spread; the lower mean cost of these variants is traced, through a cost decomposition, to the slowly learned terminal-SoC term and the shorter training budget, not to the removed term. Halving and doubling w_V and w_f move reserve, undervoltage, and terminal SoC in the expected directions (Table 11).

Changes in the manuscript: Section III-B, Section IV-L, Tables 10 and 11.

Comment 1.9

Report genuine training episodes, learning settings, convergence criteria, and evaluation on unseen conditions with multiple seeds and varying load, solar, and fleet participation.

Response: Done as described under Comment 1.1. The scenario generator varies load level, cloud cover, intra-day PV variability, fleet participation, initial SoC, and disturbance size. Test days come from an independent random stream, and the stress days use heavier load, lower participation, and larger disturbances than any training day, combined with clear-sky PV at the upper end of the training range. Results are given as mean $\pm$ standard deviation across days, with the seed-to-seed standard deviation in parentheses.

Changes in the manuscript: Sections II-C, III-C, III-E, and IV-B to IV-K.

Comment 1.10

Identify the specific technical innovation and compare with prior work.

Response: The revised contribution is the joint, learned allocation of each fleet's converter capability between hourly voltage support and frequency reserve, evaluated on a branch-resolved distribution model and a dynamic two-area LFC model at the same time. Voltage-oriented V2G studies do not model frequency dynamics, and frequency-oriented V2G studies do not evaluate feeder voltages; aggregate PEV models for primary frequency control fix the participation in advance. The results show the value of this allocation: PPO holds more reserve than the fixed-share benchmarks where vehicles are connected and inertia is low, releases it when voltage support needs the converter, and still achieves lower undervoltage than a perfect-foresight schedule. Where the gains are small, as for the frequency nadir (5% relative to the day-ahead optimization), the manuscript says so.

Changes in the manuscript: Section I (rewritten), contribution list, Section IV-J.

Additional question 4 (references). We added references on aggregate PEV frequency control models, two-time-scale RL voltage control, low-inertia systems, LFC modeling, the power-flow method, pandapower, Stable-Baselines3, generalized advantage estimation, and reproducibility in deep RL.

Reviewer 2

Comment 2.1

Train the controller, report real learning curves from several seeds, and remove Figs. 3 to 8.

Response: PPO is trained with 5 seeds; Fig. 4 shows the real learning curves on training and validation episodes. Old Figs. 3 to 8 were removed.

Comment 2.2

Explain how Fig. 12 was produced; under Eqs. (3) to (6) the minimum voltage cannot reach 0.955 p.u.; conflict with Fig. 14.

Response: The reviewer is right. The previous voltage equations and figures were inconsistent and have been withdrawn. All voltages in the new manuscript come from the AC power flow, and every voltage figure and table is generated from the same simulation records by the released scripts.

Comment 2.3

Explain the gap between Figs. 10 and 11 (SoC change inconsistent with Eq. (9)); exact zeros in Fig. 10; noisy Fig. 16.

Response: These figures were inconsistent with the previous model and have been withdrawn. In the new version, SoC follows the energy balance of Eq. (6), which can be checked against the reported set-points (Fig. 5(c) and (e)).

Comment 2.4

Use a real network or rename the environment.

Response: See the response to Comment 1.2.

Comment 2.5

Use a real two-area frequency model with inertia, damping, governor, per-area frequency, tie-line power with a synchronizing coefficient, and ACE, at a suitable time step; separate the time scales.

Response: Implemented exactly as suggested (Section II-E): per-area frequency, inertia, damping, governor and turbine, tie line with T_{12}, ACE with integral AGC, a 20-ms RK4 step within a 60-s window each hour, and an hourly scheduling layer for voltage and energy.

Comment 2.6

Give the actions physical units; clip or squash actions before env.step(); add an overvoltage penalty; consider cost and degradation terms.

Response: See Comments 1.6 to 1.8. Actions are clipped before use, an overvoltage penalty is included, and a throughput-based degradation proxy is part of the reward (its effect is shown in the ablation). We also compared clipping with squashing: a tanh-squashed Gaussian policy with generalized state-dependent exploration was trained with three seeds at the ablation budget. It removes saturation by construction but reaches a higher cost than the clipped policy at that budget (Table 7); the clipped policy is therefore retained and its saturation per action dimension is reported. An electricity-cost term was not added, because the paper is about the technical allocation of converter capability; it is listed as future work.

Comment 2.7

Use a realistic daily load profile; the $\sin^2$ *shape peaks at midnight and noon.*

Response: Correct; the previous profile had a 12-h period. The new profiles (Fig. 3) have a residential evening peak at 19:00 to 21:00 and a mixed commercial and residential daytime plateau. They are synthetic and are randomized in level and noise for each scenario.

Comment 2.8

Report voltage violation hours and magnitude, frequency deviation statistics, SoC at end of day, and energy cycled across seeds and scenarios.

Response: All of these are reported in Table 2 for the test days (Table 9 gives the main subset for the stress days, and the full stress-day records are in the released data): violation hours, integrated undervoltage and overvoltage, minimum and maximum voltage, mean and worst nadir, RMS frequency deviation, RoCoF, tie-line deviation, terminal SoC deficit, fleet throughput, unmet mobility energy, projection events, and losses, across 5 seeds and 30 test or 20 stress days.

Comment 2.9

Share the code and data.

Response: The complete code (environment, models, verification, training, evaluation, figure scripts), the trained policies, and the result files are available at `https://github.com/kenoto99/v2g-ppo-two-area` (release v3.0.1) and archived on Zenodo (DOI 10.5281/zenodo.23851585), as given in the Data and Code Availability statement.

Comment 2.10

Cut the repeated disclaimers once real results are in; fix the keywords and subject category.

Response: The disclaimers were removed because the underlying limitations were removed. Remaining limitations are stated once, in Section IV-M. The keywords now are: electric vehicles, frequency regulation, load-frequency control, proximal policy optimization, reinforcement learning, vehicle-to-grid, voltage regulation.

Additional question 1. The previous evaluation was a single seeded rollout of an untrained policy. The new results are from trained policies evaluated on unseen days.

Additional changes made after a technical review of the revised manuscript

Before resubmission the revised manuscript was examined by an independent technical reviewer, whose eight comments are addressed below. The analyses added in response reuse the trained policies and the test-day scenarios; no training run was repeated. The author list was also updated with the agreement of all authors: Wasan Phetphimoon is no longer an author, and Kritphon Phanrattanachai (Phetchabun Rajabhat University) was added as the sixth author.

Comment A.1

Fairness of the comparison with DAO: PPO adjusts both the V2G power and the reserve share, whereas DAO fixes the share at 0.3 and does not optimize the frequency part.

Response: We agree that the fixed share could have favored PPO. The benchmark was therefore re-optimized for every share $\rho \in \{0, 0.1, \ldots, 0.6\}$ on the same test and stress days, and a per-day oracle (the share with the lowest realized cost on each day, an upper bound on any tuned fixed share) was added. The fixed share used before was close to, but not at, the best fixed share ($\rho = 0.4$); PPO remains 10.4% below the best fixed share and 8.8% below the oracle on the test days (lower on 30 of 30 days, $p = 1.9 \times 10^{-9}$) and 4.3% below it on the stress days. The sweep also shows the trade-off that no fixed share resolves (nadir improves and undervoltage worsens with ρ), which PPO avoids by moving the share hour by hour. The structural difference between the two controllers (DAO's objective excludes the frequency terms) is now stated explicitly.

Changes in the manuscript: Section III-D (last paragraph); new Section IV-D and Table 4; Conclusion.

Comment A.2

The daily cost of Table 2 is a weighted sum of technical penalties, not an economic cost; its meaning and units should be stated before relative reductions are interpreted.

Response: The quantity is now defined as the daily performance cost, a dimensionless weighted penalty with no price, reserve payment, or degradation cost, in Section III-B; the table row, the learning-curve axis, the first sentence of Section IV-C, and the abstract say so, and the relative reductions are described as relative technical penalty under the chosen weights.

Changes in the manuscript: Section III-B (new last paragraph); abstract; Section IV-C; row label of Tables 2 and 9; Fig. 4.

Comment A.3

The voltage results do not meet the band in every hour (PPO minimum voltage 0.925 p.u., 11.9 violation hours per day); this should be presented as a reduction of severity, not a guarantee.

Response: Agreed. The voltage paragraph of Section IV-C now states that the results are a reduction in depth and duration of undervoltage and not compliance in every hour, gives the number of area-hours that remain outside the band, and explains why the feeder itself sets the floor (nominal Baran–Wu minimum of 0.9131 p.u. with no EVs; the smallest plugged-in fleet during the evening peak). The abstract and the conclusion carry the same qualification.

Changes in the manuscript: Section IV-C, paragraph *Voltage*; abstract; Conclusion.

Comment A.4

Terminal SoC deficit (6.31% on test days, 18.02% on stress days): explain the consequence for driving energy and consider a hard terminal constraint or consecutive-day tests.

Response: The deficit is now translated into energy per vehicle (1.9 kWh, about 10 km or 21% of a day's driving energy on the test days), and the unmet mobility energy of the day is reported alongside. More importantly, the 30 test days were run consecutively with the terminal energy carried over (new Table 3). Costs rise for all controllers, and the residential fleet drifts to a lower SoC over the month with all three; PPO degrades least (unmet mobility energy

13.9 MWh on 8.2 days, against 108.8 MWh on 24.0 days for DAO), but the drift shows that the soft terminal penalty does not guarantee a sustainable energy state. We state this plainly, identify the hard terminal constraint (constrained RL or a projection at the last plugged-in hours) as the remedy, and did not implement it in this revision.

Changes in the manuscript: Section IV-C, paragraph *Energy and fleet*; new Table 3; Conclusion; Supplementary S6 and S7.

Comment A.5

Limits of the aggregated EV model: how are parked and driving vehicles separated, and how does the single-SoC assumption affect deliverable power?

Response: Section II-D now explains that the aggregation separates parked and driving vehicles in power (through the plugged-in fraction) but not in energy (driving energy is withdrawn from the common store while vehicles are away), that all plugged-in vehicles are assumed to share the fleet-average SoC, and that the energy-feasible limits are exact for the aggregate but optimistic for the deliverable power of a dispersed fleet near its SoC limits, most of all at the end of the evening discharge. A vehicle-level allocation layer is listed as future work.

Changes in the manuscript: Section II-D; Section IV-M item (iii); Supplementary S7.

Comment A.6

Coupling between the frequency reserve and the feeder power flow: when reserve is activated, do feeder voltages and currents stay within limits?

Response: This was not checked before and the check changes the picture. Every test-day hour was re-solved at $P^{\text{set}} \pm R$ and at the realized peak droop power (new Section IV-G and Table 8). Upward activation helps the voltage; full downward activation (the fleet absorbing the whole reserve) would take 32.5% of the PPO area-hours that are inside the band at the schedule out of it, with a minimum voltage of 0.884 p.u. The simulated disturbances never reach that level: the realized peak droop power averages 15.7% of the committed reserve, at which 9.6% of the area-hours leave the band for a few seconds, with an additional drop of at most 0.012 p.u. Branch currents change little and the case data carry no thermal rating. The exposure is small for the modeled disturbance class but real and growing with the committed reserve, so we report it as a limitation and identify a voltage-aware (asymmetric) reserve rule as the remedy.

Changes in the manuscript: New Section IV-G and Table 8; Conclusion; Supplementary S6.

Comment A.7

Effect of raw-action saturation (49.2%) on exploration, gradients, and transfer; limits of the squashed-policy comparison.

Response: Section IV-F now discusses the three effects separately (exploration collapses to the boundary action, the likelihood-ratio update keeps pushing the mean outward with no corrective signal from the environment, and a saturated output is benign only while the deployed limits coincide with the training limits) and states the limits of the squashed-policy comparison (three seeds, half budget, different exploration mechanism, no re-tuning, one parameterization), so that it is read as "the bounded variant we tried did not help under equal conditions" rather than as a general statement.

Changes in the manuscript: Section IV-F, two paragraphs.

Comment A.8

Realism and scope: discuss forecast errors, communication delay, charger heterogeneity, converter dynamics, and parameter uncertainty.

Response: The Limitations section now has a dedicated paragraph with these five items, each with its consequence for the reported results and the step needed before field use (training on forecast errors, delays, and parameter ranges; a vehicle-level dispatch layer; a fuller converter model). The stress days are explicitly described as varying load, participation, and disturbance level but not forecast accuracy.

Changes in the manuscript: Section IV-M (new paragraph, items (i) to (v)); Conclusion; Supplementary S7.

We thank the reviewers again. Their comments led to a substantially stronger study.

Sincerely,
Krischonme Bhumkittipich, on behalf of all authors